What is Cryptography Hiding from Itself?

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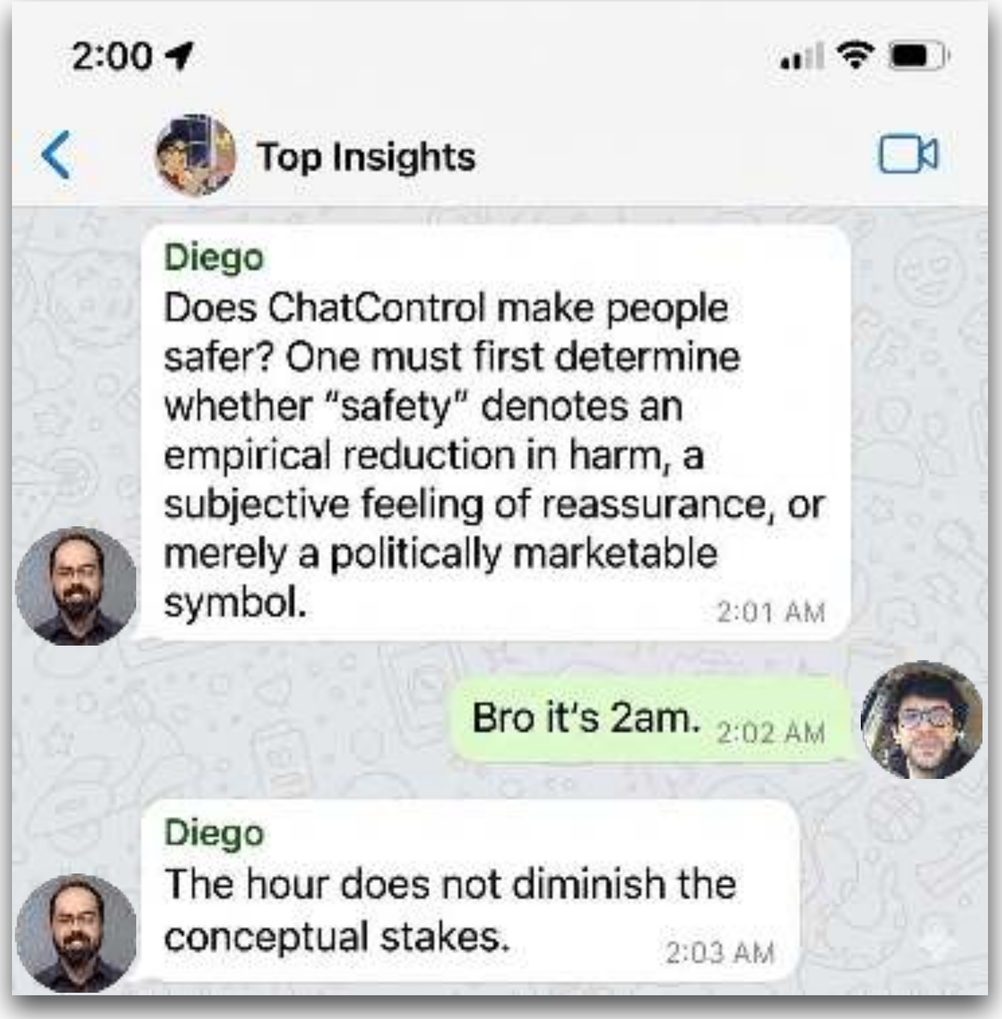
A starting point

“Cryptography rearranges power.”

— Phillip Rogaway, 2015

Top Insights



What E2EE means

Formally

The service provider can't read my ciphertext in transit

Socially

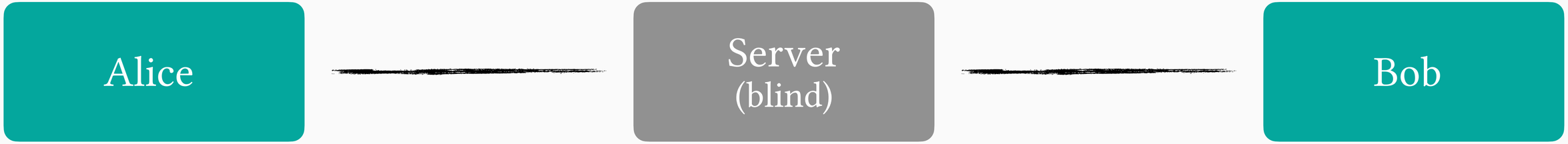
The channel is not built to routinely inspect my private life.

ChatControl — Fast context

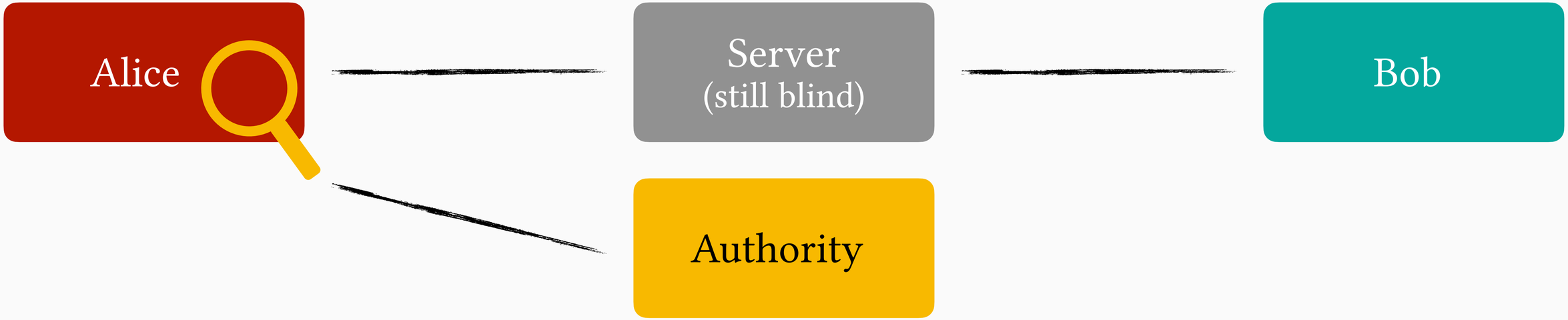
Proposal	EU regulation: detect, report, remove CSAM across digital services.
Scope	Technology neutral — does not exempt end-to-end encrypted services.
Obligation	Expressed as outcome: “achieve detection”.
Implication	If you keep E2EE and must detect, inspection moves to the endpoint.

Where does inspection happen?

Without CSS



With CSS



Two worlds of detection

Hash Matching

Known CSAM

Classifier-based

Unknown content

General monitoring is already rejected

"...cannot be regarded as necessary in a democratic society."

— ECHR, Podchasov V. Russia, 2024

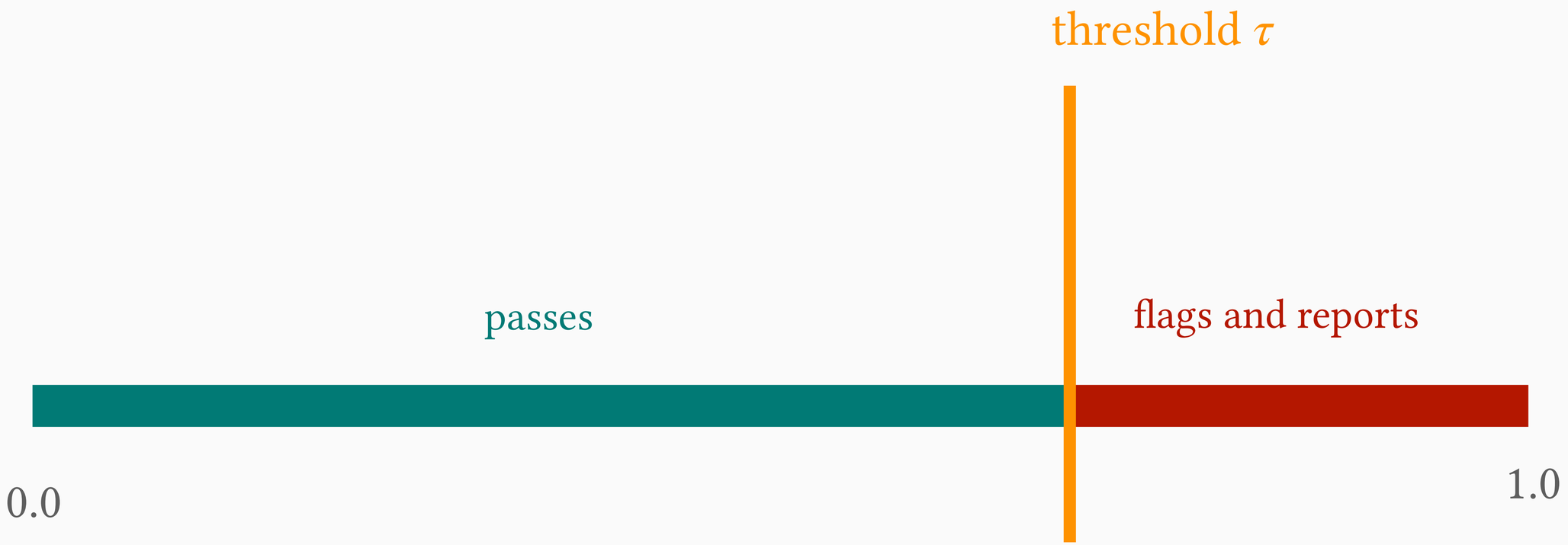
Pivot

Up to now: plumbing

From here: something stranger and more important

ChatControl as a primitive

The threshold is not a footnote



Quick show of hands

Lower τ

Catch more.

Higher τ

Protect more innocents.

Safety becomes an epistemic construct

Political question:

“What harms are intolerable?”

→

Technological question:

“What signals exceed a detection threshold?”

When measurement replaces deliberation

“Technologies of hubris” are governance tools that try to render uncertainty manageable through prediction, calculation, and claims of control.”

— Sheila Jasanoff
(roughly)

The paradox of protection

Negative liberty

Freedom from interference.

The right to be left alone.

Positive liberty

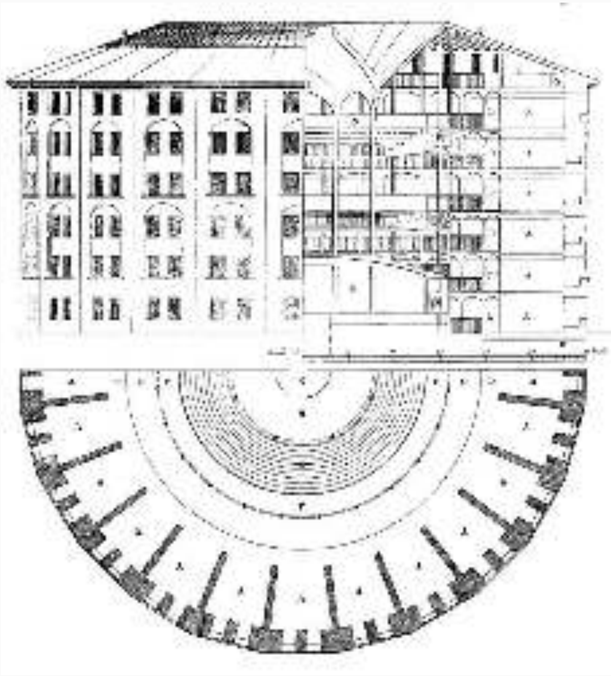
Freedom to pursue safety, collective goods, protection.

Artifacts have politics

Technologies can be understood as forms of order as their architectures predispose certain social and political outcomes.”

— Langdon Winner
(roughly)

The Panopticon



Individuals know they may be watched,
even if they aren't continuously observed.

That uncertainty produces self-discipline.

Undone science

“Some questions are not just unanswered; they are marginalized by how research agendas, funding systems, and institutions decide what gets studied.”

— David Hess
(roughly)

Four questions the system doesn't ask

Scale

False positive volume at EU population scale.

What does remediation look like for innocents?

Adversaries

Once attackers learn the scanner,

what is the equilibrium?

Security regressions

Scanning logic + update channel

on billions of endpoints = new attack surface

Governance

Who controls model updates,

DB membership, auditing, and appeals?

Failure modes

Scale

Base rates → false alarms dominate operational reality

Security

Endpoint scanning + distribution/update channel = attack-surface magnet.
New exploit classes.

Creep

Once the inspection primitive exists,
it becomes cheap to reuse for new categories.

Our community's comparative advantage

Cryptographers are unusually good at predicting second-order failures.

If we can name the failure modes clearly, we can help policy avoid building brittle infrastructure.

“Ok doomer.”

— You *(probably)*

Rejecting ChatControl as drafted ≠ doing nothing

Targeted warrants

Judicial authorization,
individualized suspicion

Better coordination

Resources, training,
cross-border ops

Takedown at source

Cut supply —
not inspect transit

Opt-in safety features

User/family control,
not state mandate

Build architectures of protection
that do not demand obedience
as their price.

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Check out https://eprint.iacr.org/2025/1951

Thanks